Note on provided yolo model

The provided model “yoloe-26n-seg-strawberry.onnx” is a version of the YOLOE-26n-seg model. The YOLOE series take a list of text labels as input on initialisation. In this case, I have saved a version with the weights “frozen” to only detect the class “strawberry”. The model can detect objects and also provide segmentation masks for them. Some more details:

- The input “blob” size is expected to be 640 by 640 pixels.
- The other normalisation values from workshop 3 example seem to work fine.
- The network outputs two tensors of values. You can get the names of the outputs using *getUnconnectedOutLayersNames()* of the dnn (should be ('output0', 'output1')). This list of names can be passed to the dnn *forward* method to get the list of outputs.
- The first output (‘output0’) has shape (1, 300, 38) – this gives 300 detections, each with an associated vector of length 38. The first 4 values (elements 0 to 3) of this vector give the bounding box coordinates. The next value (element 4) gives the detection confidence, the next value (element 5) gives the class (which is always ‘strawberry’), and the next 32 values (elements 6 to 37) give the weights for the segmentation mask of the object.
- The second output (‘output1’) has shape (1, 32, 160, 160) and contains 32 mask ‘prototype’ images of size 160x160. For this you will need to combine the 32 weights from ‘output0’ with the 32 prototype masks from ‘output1’. Each weight multiplies the corresponding prototype mask, and the results are summed i.e.

$$M(x,y) = \sum_{i=0}^{31} w_i P_i(x,y)$$

  Where $M(x,y)$ is the mask, w_i is the weight for prototype image i,and $P_i(x,y)$ is the i^{th} prototype image. This calculation is best done using numpy operations for efficiency (https://numpy.org/doc/stable/reference/).
- Once the mask is calculated it needs to be resized to the image size. The values in the mask are floating point numbers, and were trained with 0 for background and 1 for the foreground, but the resulting mask can include values outside of this range, including negative numbers. A suitable threshold needs to be applied to select the strawberry’s pixels.